

DSA Problem Ladder — Day 3

Wednesday — Hard: derive a solution from familiar parts

September 16, 2026 | Priyanka | 120 minutes | C11 or Python 3

Central question: can a routine compute more than its obvious output, or search an answer that is not stored in an array?

1. Count out-of-order pairs. For up to 200,000 integers, count pairs $i < j$ with $a[i] > a[j]$. Equal values do not count. Require $O(n \log n)$ time and at most $O(n)$ auxiliary space. Example: $[3, 1, 2, 1] \rightarrow 4$. Start with a small brute-force checker; explain how the faster algorithm counts every pair exactly once. Use a 64-bit count in C.
2. Smallest daily capacity. Positive integer parcel weights must be shipped in their given order over at most D nonempty days, where $1 \leq D \leq n$. Each day's parcels form a contiguous block. Find the smallest integer capacity allowing all parcels to ship. Require $O(n \log(S+1))$ time, where S is the sum of weights, and $O(1)$ auxiliary space. Example: $[3, 2, 2, 4, 1, 4]$, $D=3$ gives 6. Include $D=1$, $D=n$, and one dominant weight; use wide sums in C.

Session rules

- Minutes 0–10: closed-notes recall of one invariant and an edge case from the previous session (Monday: last week's boundary search).
- Minutes 10–110: solve today's queue in order. Check time after roughly 25 minutes per problem on easy days, 33 on medium days, and 50 on Wednesday. Unused time transfers forward; the queue is not a quota.
- Minutes 110–120: stop coding, record results, and explain one failed assumption and its correction. Mark unfinished work partial at the cutoff.
- Use C11 OR Python 3 consistently throughout the timed week. Translation is optional and does not count as another solve.
- Before coding, state a direct approach, its cost, and what property might remove work. After coding, give a correctness argument, time/space costs, and normal, boundary, and adversarial tests.
- After 15 minutes without a useful new step, write the blocker and choose to continue, request a small hint, or park the problem. Record hints; keep complete solutions closed during the attempt.

Count a solve only when working code meets the stated constraints, passes checks, and you can explain why it works. Separate independent and assisted solves. A correct slow baseline is partial if it misses the required bound. Testing time counts toward the session. If finished early, improve tests and proofs without increasing the solved count.

Attempt record

For each problem, submit its number, minutes spent, independent/assisted/partial/parked status, hints used, code, test results, invariant, and time/space analysis. Keep the same language throughout the timed week.

At 120 minutes record: attempted __; independently solved __; assisted solves __; partial __.

Exit ticket: Which assumption or boundary was hardest? Give one smallest failing example and explain your correction. Counts are evidence of this session, not a quota.